



IEEE Testbed UI Manual

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1. Abbreviations

Abbreviation	Definition
API	Application programming interface
IMEI	International Mobile Equipment Identity
IMSI	International Mobile Subscriber Identity
MCC	Mobile Country Code
MNC	Mobile Network Code
NGAP	Next Generation Application Protocol
PCAP	Packet Capture
PDU	Protocol Data Unit
SUPI	Subscription Permanent Identifier
TAC	Tracking Area Code
UE	User Equipment
UI	User Interface

2. Introduction

IEEE 5G/6G Innovation Testbed is an end-to-end cloud based 5G network. Testbed resources can be accessed using CLI and UI modes. This document describes the accessing of Testbed using UI mode.

3. About Testbed UI

IEEE Testbed UI is a JavaScript based Graphical User Interface for IEEE 5G/6G Innovation Testbed. It exposes the Testbed capabilities through APIs for simplifying testbed operations.

Design:

- Frontend: Angular JS
- Backend: Node JS/Express JS
- Database: MongoDB

3.1. Testbed UI Sections

IEEE Testbed UI contains various sections (given below) for accessing and managing each entity of the Testbed.

- Dashboard
- Users
- Testbed
- User cases & Tools
- Resources

3.2.1 Dashboard

Dashboard defines the overall network overview of testbed including the gNodeB and 5G core network function. It also shows the below major KPIs that defines the overall status of the testbed

- Network Overview for gNodeB and 5G core.
- KPI(s) for Users, UE(s), Monitoring and Tracing.
- Use cases and tools information

3.2.2 Users

This section contains the user management details.

- **Admin:** Can add, modify and delete users
- **Non admin users:** Can view the configured details and download the required key files (for ssh login).

3.2.3 Testbed

This section contains management of gNodeB, UEs, 5G core, Tracing and Monitoring of the Testbed.

3.1.3.1 gNodeB

gNodeB is the 5G wireless base station that transmits and receives messages between the UE and the 5G core network. gNodeB establishes NGAP connectivity with the 5G Core.

- Admin can update the configuration for gNodeB (Mainly “NGAP IP address”, “GTP IP address” and “AMF IP address and port”)
- Admin can start/stop gNodeB.
- Non-admin users can view gNodeB details and can start gNodeB.

3.1.3.2 UE

Users can create emulated UEs (User Equipment) and perform operations in this section.

- UEs can be added by pressing “Add UE” button.
- Once UEs are created, users can “Modify”, “Delete” or “Register” UEs in the “Configuration” tab.
- Registered UEs will appear in “Operations” tab.
- In the “Operations” tab, users can perform below options on UE.
 - View UE’s data (like IMSI, IMSI, status, timers, tunnel information etc.)
 - Establish PDU session
 - PDU session release
 - Deregister
 - Release all PDU sessions
 - Dump all UE information (to a file)

3.1.3.3 5G core

The 5G Core in the Testbed is orchestrated using Kubernetes. Each network function of the 5G core is represented as a Kubernetes Pod.

In this section, users can view various aspects of the 5G core (given below) and configure subscriber information.

- System metrics (CPU Usage and Memory Usage)
- PODs List (network functions)
- Services List
- Replica List
- Deployment List
- Config Maps List
- In “Logging” tab, users can view and download the logs.

3.1.3.4 Monitoring

This section provides the capability to the users for monitoring 5G core metrics using Grafana and 5G core network functions using tools like Linkerd and Kubernetes dashboard.

3.1.3.5 Tracing

This section provides the capability to the users for capturing PCAP logs for the test scenarios.

3.2.4 Use cases and Tools – Under development

This section provides various tools and provisions for different uses cases. Below is the list of tools and use cases supported in this section.

- CAMARA API Integration.
- Network Slicing
- Linux Traffic Control Integration.
- Open Speed Test
- Trace Analyzer
- DDoS Attack

3.2.5 Resources

This section provides resources for different uses cases and features available in the Testbed. Resources include the list of documents and videos for different uses cases and features

4. Accessing Testbed UI

This section explains the steps required for accessing the IEEE Testbed UI.

Step-1: Connect to UERANSIM EC2 instance using forwarding method.

IEEE Testbed UI runs on the standard port 4200 for frontend and 8400 for backend support.

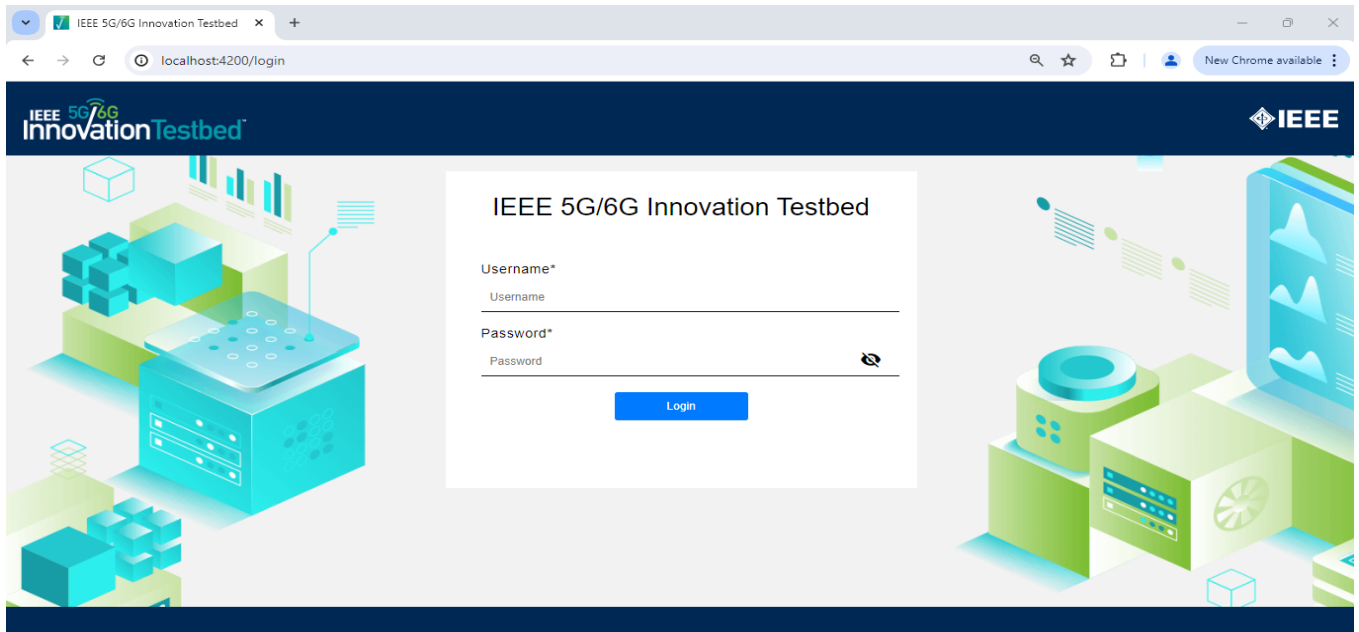
For accessing UI, users need to connect to UERANSIM EC2 instance using the port forwarding method.

- `ssh -I <key_file> username@<ip-address> -p 3022 -L 4200:127.0.0.1:4200 -L 8400:127.0.0.1:8400`

Example: `ssh -i "ieee-aws-ssh-key.pem" ubuntu@13.233.101.116 -p 3022 -L 4200:127.0.0.1:4200 -L 8400:127.0.0.1:8400`

Step-2: Login to UI in any browser using the below URL.

- `http://localhost:4200/login`

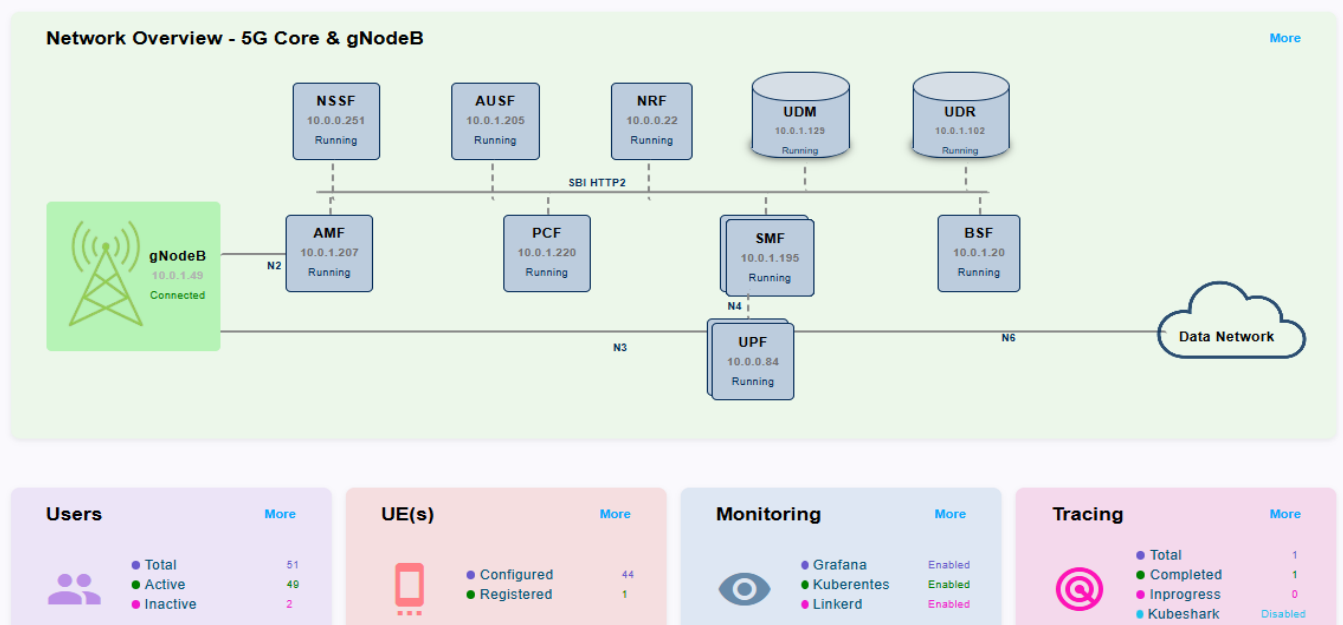


Once a user enters the credentials (username and password), he or she will be redirected to the dashboard page

4.1 Dashboard

This section defines the dashboard containing overall network overview of testbed including the gNodeB and 5G core network function, KPI(s) for Users, UE(s), Monitoring, Tracing & Use cases and tools information

Testbed Dashboard



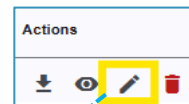
2. IMSI numbers should be prefixed with “imsi-”.
 3. IMSI numbers/ranges should not overlap with IMSI numbers/ranges of other users.
- Number of UEs: Value provided in this filed is used to create UEs for the users starting from IMSI number specified in the filed “Starting SUPI”.
 - Password

Add User

User Name*	user1	Email*	user1@ieee.org
First Name*	Test	Last Name*	User
Starting SUPI*	imsi-208930000000201		
Num Of UE's*	10		
Password*			
Submit Cancel			

4.2.2 Editing User details

Admin can edit user’s password by clicking edit button under “Actions” column.



Edit User

User Name	user1	Email	user1@ieee.org
First Name	Test	Last Name	User-1
Starting SUPI*	imsi-208930000000201		
Num Of UE's*	10		
Status	☒ Active ☐ Inactive		
User ID	679ff49f372ae24114576935		
Select a password for user			
Password*			
Confirm Password*			
Update Cancel			

4.3 Testbed management

In this section, users can get information about accessing gNodeB, UE, 5G core, Tracing and Monitoring of the Testbed.

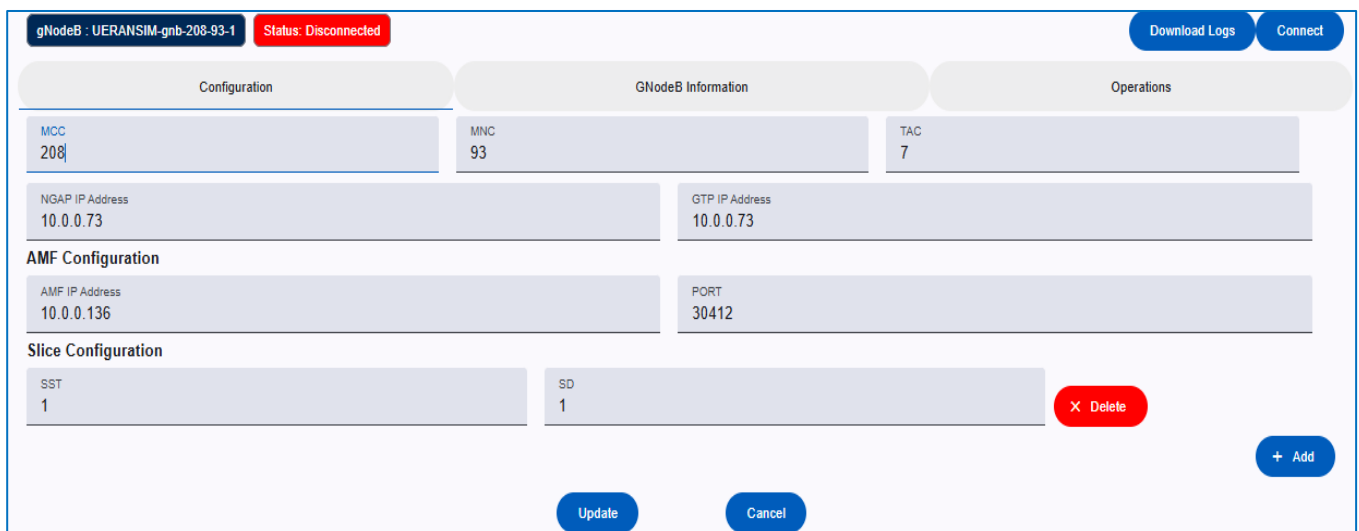
4.2.1 gNodeB

This section contains the sub-sections below.

- Configuration
- Information
- Operations

4.3.1.1 Configuration

Admin can update “MCC”, “MNC”, “TAC”, “NGAP IP Address”, “GTP IP Address”, “AMF IP Address”, “Port” and Slice configuration.



The screenshot shows the gNodeB configuration interface. At the top, it displays 'gNodeB : UERANSIM-gnb-208-93-1' and a red 'Status: Disconnected' button. There are 'Download Logs' and 'Connect' buttons in the top right. The interface is divided into three main sections: Configuration, GNodeB Information, and Operations. The Configuration section contains fields for MCC (208), MNC (93), TAC (7), NGAP IP Address (10.0.0.73), GTP IP Address (10.0.0.73), AMF IP Address (10.0.0.136), and PORT (30412). The Slice Configuration section shows a table with columns SST and SD, containing the value 1. There is a red 'Delete' button next to the slice entry and a blue '+ Add' button at the bottom right. At the bottom center, there are 'Update' and 'Cancel' buttons.

4.3.1.2 Information

This section will be updated when gNodeB is in connected state.

gNodeB : UERANSIM-gnb-208-93-1
Status: Connected
Download Logs
Connect

Configuration
GNodeB Information
Operations

NCI : 16
PLMN: 208/93
TAC: 7
NGAP IP: 10.0.0.73
GTP IP: 10.0.0.73
PAGING DRX: v128
IGNORE SCTP ID: true

ID	Name	Address	State	Capacity	Association	Served GUAMI	Served PLMN
2	open5gs-amf0	10.0.0.136:30412	CONNECTED	255			

SST	SD
1	1

UE ID	RAN NGAP ID	AMF NGAP ID
No records found		

Association Details

ID: 2
RX Number: 10
TX Number: 10

Close

Served GUAMI Details

PLMN: 208/93
Region ID: 2
Set ID: 1
Pointer: 0
Backup AMF: None

Served PLMN

PLMN 208/93
NSSAI
SST: 1, SD: 1
SST: 1, SD: N/A
SST: 2, SD: N/A
SST: 3, SD: N/A
SST: 1, SD: 2
SST: 1, SD: 3
SST: 1, SD: 4

4.3.1.3 Operations

Various gNodeB operations are supported in this section.

UE Release: Release NGAP connection of the UE

Disconnect: gNodeB will be disconnected from 5G core and all UEs will be deregistered (if any).

Download gNodeB Information: gNodeB Information will be downloaded to a local file.

gNodeB : UERANSIM-gnb-208-93-1
Status: Connected
Download Logs
Connect

Configuration
GNodeB Information
Operations

UE Release
Disconnect gNodeB
Dump all UE Information

UE Release

UE ID

Cancel
Proceed

Disconnect gNodeB

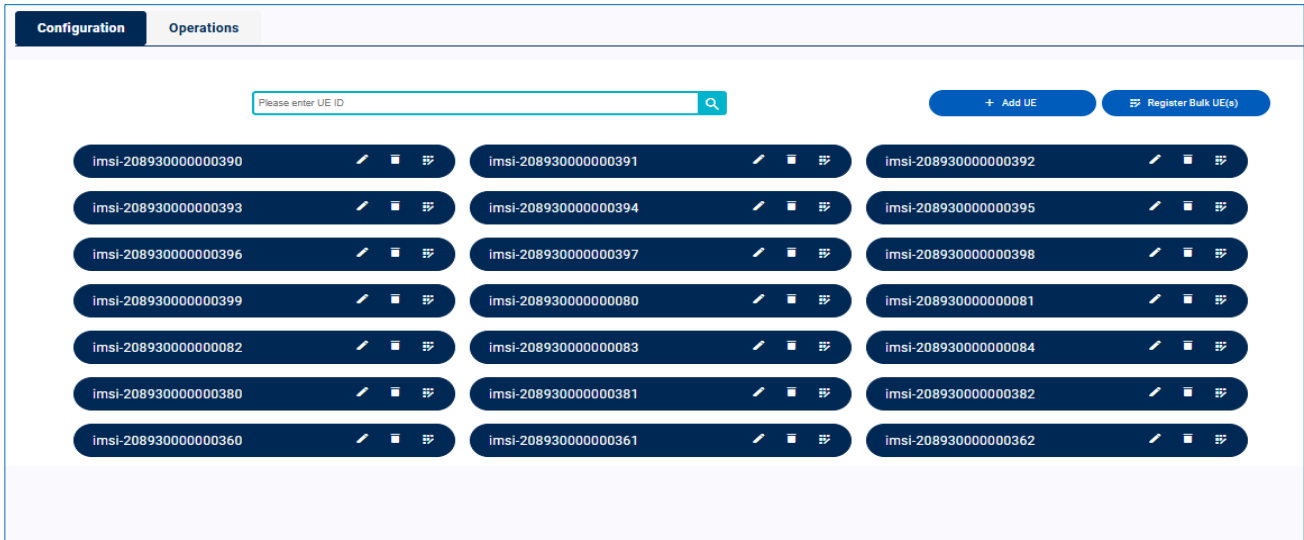
On disconnecting gNodeB, all the UE(s) will get deregistered for all users. If you are sure, then please click on Proceed else Cancel

Cancel
Proceed

4.3.2 UE

This section contains the list of UEs configured for a particular user.

4.3.2.1 Configuration



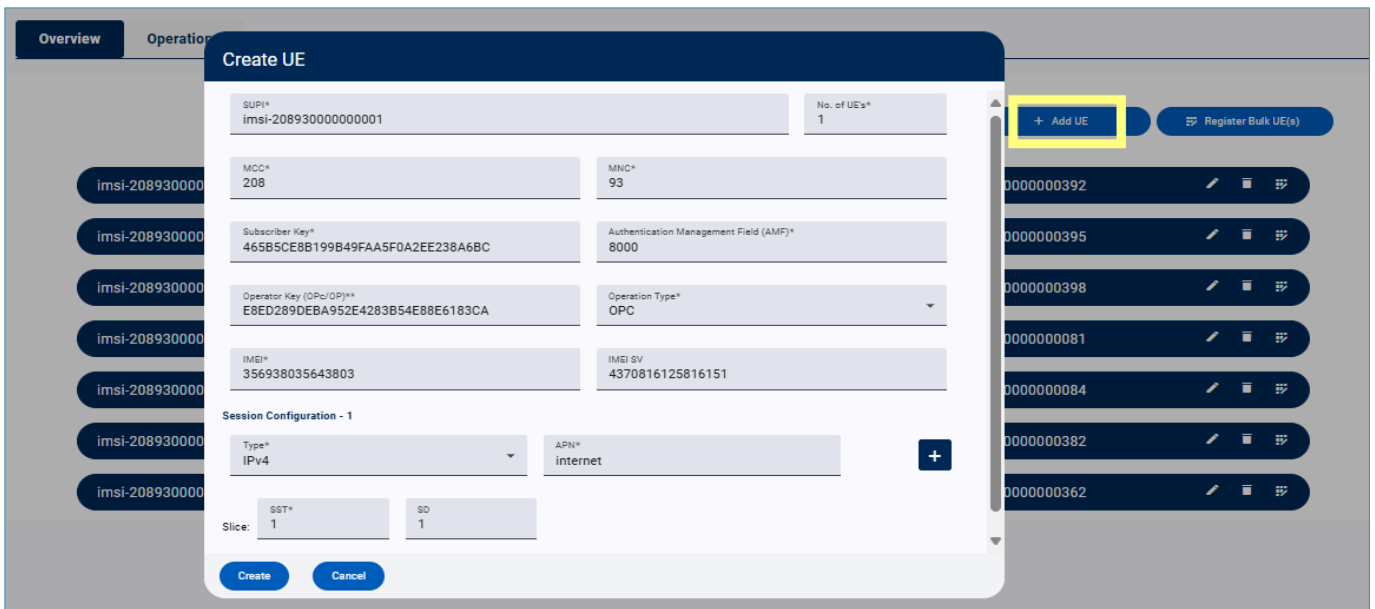
The screenshot shows the 'Configuration' tab of the UE management interface. At the top, there is a search bar labeled 'Please enter UE ID' and two buttons: '+ Add UE' and 'Register Bulk UE(s)'. Below these, a grid displays 21 UEs, each with an IMSI number and three action icons (edit, delete, and a third icon). The UEs are arranged in three columns of seven, with the last column containing only one UE.

IMSI	IMSI	IMSI
imsi-208930000000390	imsi-208930000000391	imsi-208930000000392
imsi-208930000000393	imsi-208930000000394	imsi-208930000000395
imsi-208930000000396	imsi-208930000000397	imsi-208930000000398
imsi-208930000000399	imsi-208930000000080	imsi-208930000000081
imsi-208930000000082	imsi-208930000000083	imsi-208930000000084
imsi-208930000000080	imsi-208930000000081	imsi-208930000000082
imsi-2089300000000360	imsi-2089300000000361	imsi-2089300000000362

4.3.2.1.1 Add UE

User can add emulated UEs using “Add UE” button.

The “SUPI” field should be added with the allocated IMSI number (by Admin in User section) and “No. of UE’s” can be updated with required number of UEs. Rest of the fields can leave with default values.



The screenshot shows the 'Create UE' modal form. The 'SUPI*' field is populated with 'imsi-208930000000001' and 'No. of UE's*' is set to '1'. The 'MNC*' field is '208' and 'MNC*' is '93'. The 'Subscriber Key*' is '465B5CE8B199B49FAA5F0A2EE238A6BC' and 'Authentication Management Field (AMF)*' is '8000'. The 'Operator Key (OPc/OPj)*' is 'E8ED289DEBA952E4283B54E88E6183CA' and 'Operation Type*' is 'OPC'. The 'IMEI*' is '356938035643803' and 'IMEI SV' is '4370816125816151'. The 'Session Configuration - 1' section shows 'Type*' as 'IPv4' and 'APN*' as 'internet'. The 'Slice' section shows 'SST*' as '1' and 'SD' as '1'. The 'Add UE' button is highlighted in yellow.

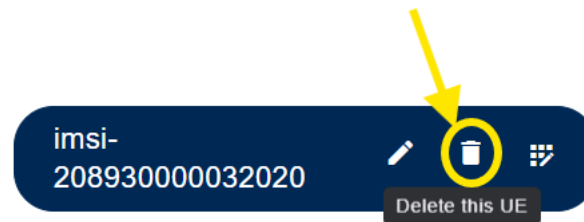
4.3.2.1.2 Edit UE

User can update the UE details using the “Edit” button.



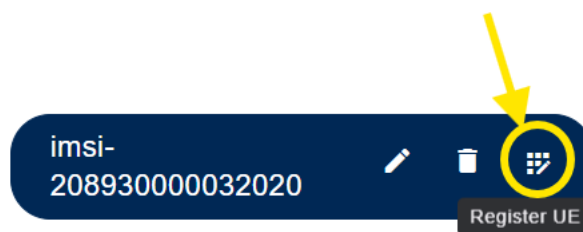
4.3.2.1.3 Delete UE

User can delete UE entry using the “Delete” button.

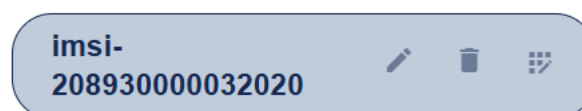


4.3.2.1.4 Register UE

Users can register the UE to 5G network by using the “Register UE” button.



Once UE is registered successfully, “Edit”, “Delete” and “Register” buttons gets deactivated, and color is changed to indicate that the registration is successful. Also, for every registered UE, an entry will be created in the “Operations” section.

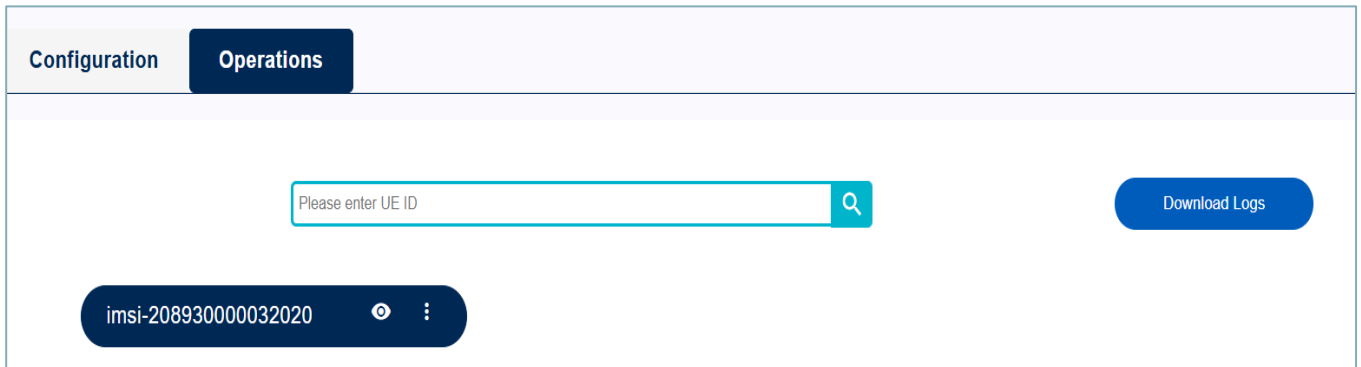


! For successful registration, UE details should be configured in the 5G Core.

Please refer to section [3.2.3](#) for configuration of subscriber details in the 5G core.

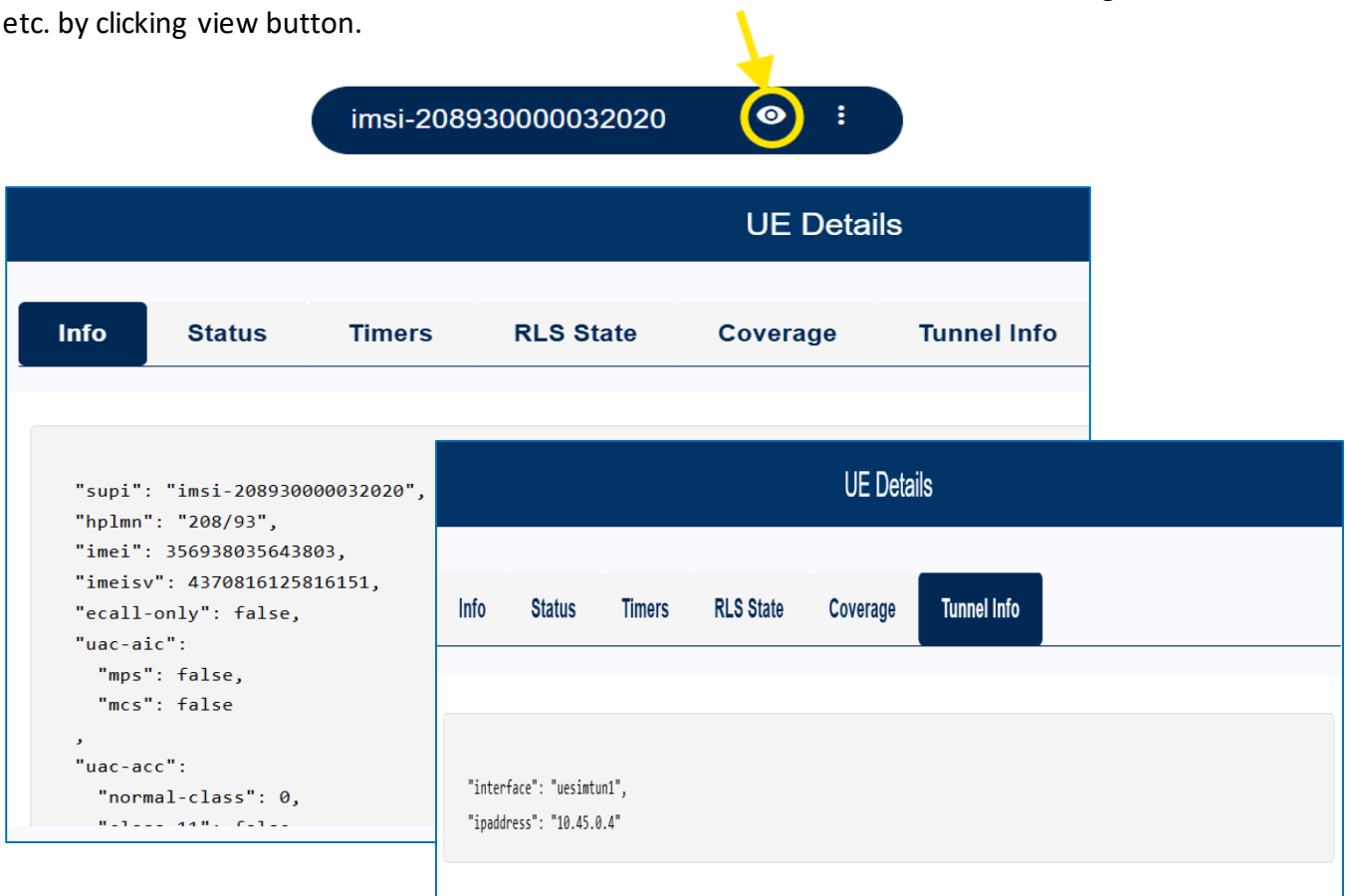
4.3.2.2 Operations

Registered UEs will appear in this section. Users can perform various operations like PDU session creation, PDU session release, De-register on the registered UEs.



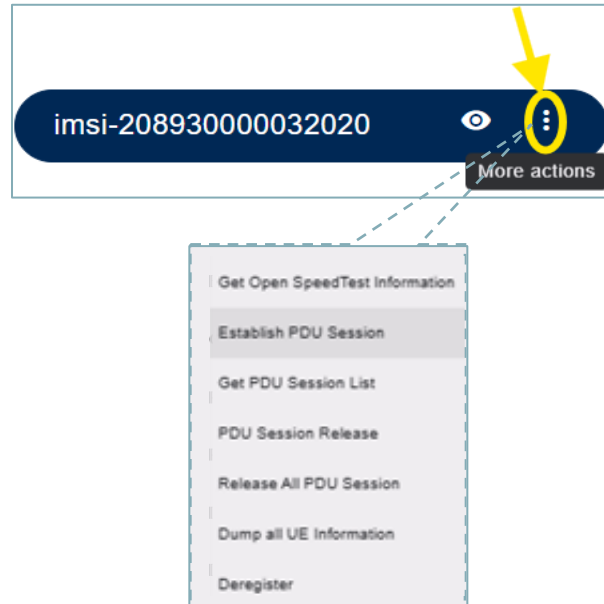
4.3.2.2.1 View Information

Users can view various information of the UE like PLMN, IMEI, Status, Timers, Coverage, Tunnel details etc. by clicking view button.

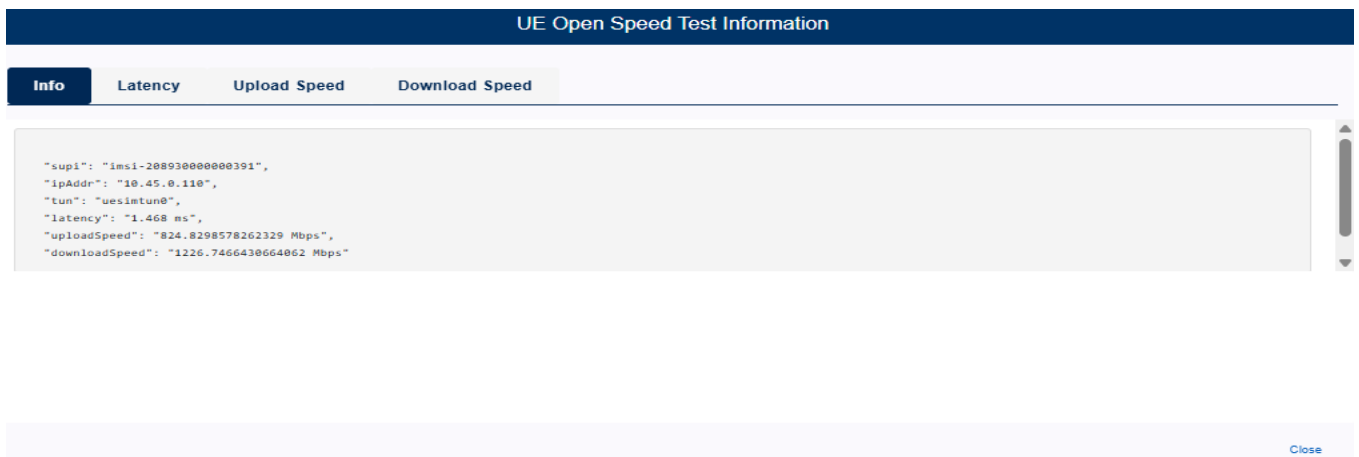


4.3.2.2.2 More actions

User(s) can perform various operations on the registered UEs using “More actions” button.



Get Open Speed Test Information: Returns the Uplink and Downlink speed for that UE.

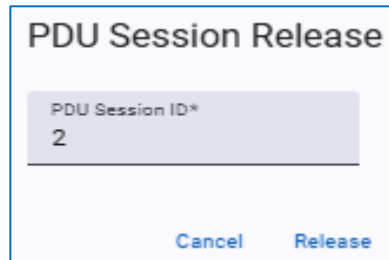


Further detailed information can be seen for Latency, Uplink speed and Downlink Speed.

Establish PDU Session: Creates the secondary PDU session

The image shows a dialog titled 'Establish PDU Session'. It contains three input fields: 'SST*' with the value '1', 'SD*' with the value '1', and 'DNN*' with the value 'internet'. At the bottom of the dialog are two buttons: 'Cancel' and 'Establish'.

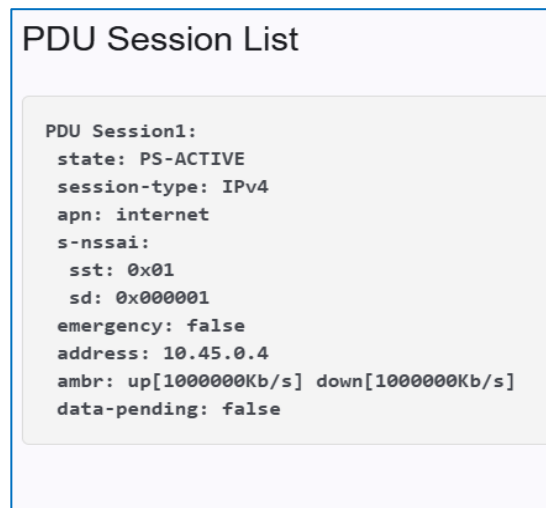
PDU Session Release: Release a particular PDU session.



A dialog box titled "PDU Session Release". It contains a text input field labeled "PDU Session ID*" with the value "2" entered. At the bottom, there are two buttons: "Cancel" and "Release".

Release All PDU sessions: Release all the active PDU sessions for this UE.

Get PDU Session List: List all the active PDU sessions of the UE



A dialog box titled "PDU Session List". It contains a text area with the following JSON-like output:

```
PDU Session1:
state: PS-ACTIVE
session-type: IPv4
apn: internet
s-nssai:
  sst: 0x01
  sd: 0x000001
emergency: false
address: 10.45.0.4
ambr: up[1000000Kb/s] down[1000000Kb/s]
data-pending: false
```

Dump all UE Information: Download all the information of a registered UE in a local text file.

Deregister: Perform de-registration of the UE with the 5G Core. UE entry will be removed from the operations section and all buttons will be enabled for this UE in configuration section.

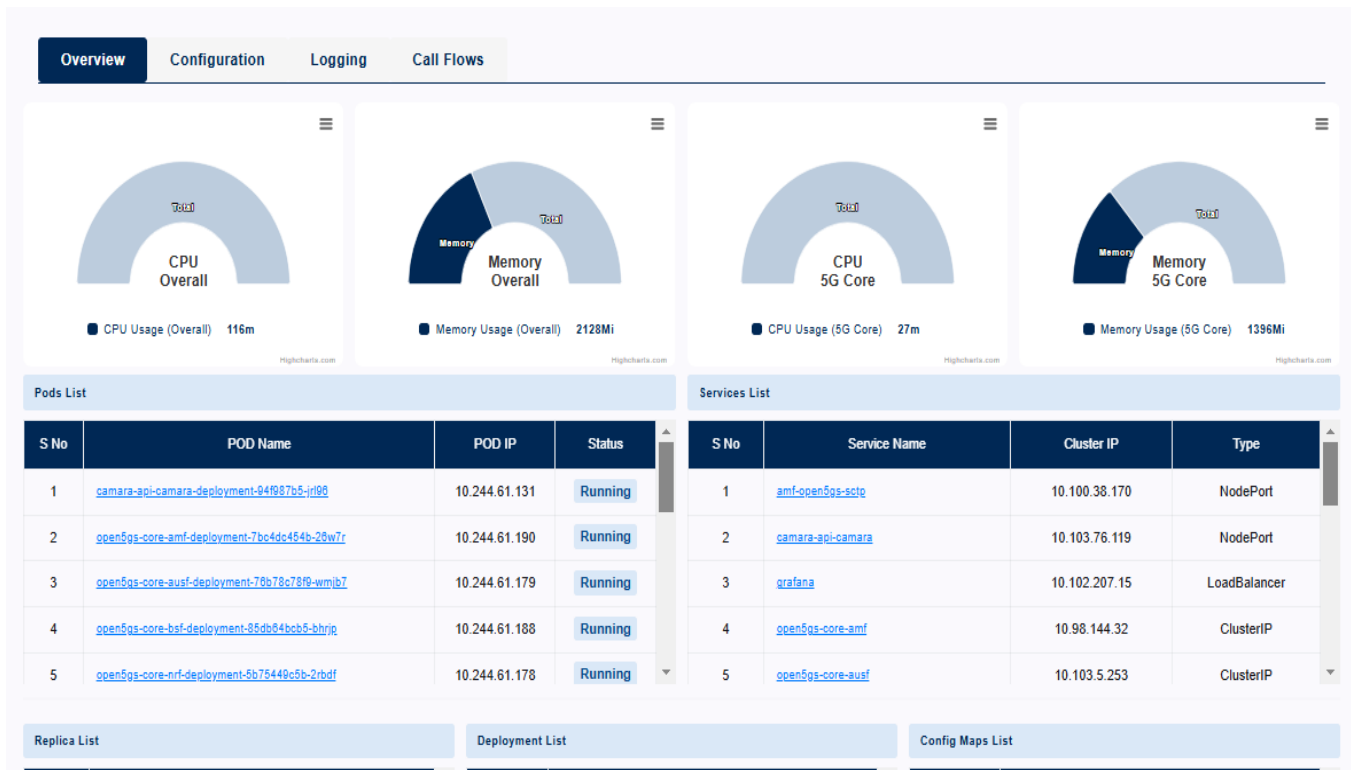
4.3.3 5G Core

This section contains the sub-sections below.

- Overview
- Configuration
- Logging
- Call Flows

4.3.3.1 Overview

This section provides an overview of the 5G core network including memory usage, CPU usage, pods list, services, deployments and config maps etc.



Each 5G network function is orchestrated as Kubernetes pod. Users can view more details of each Kubernetes pod by clicking the pod name.

4.3.3.2 Configuration

This section lets users configure the subscribers in the 5G core. Users are provided with three options for the subscriber configuration.

Option-1: “Add Subscriber configuration using Add”

- Useful for adding subscribers in bulk.
- Users need to specify the IMSI number, Key Value, OPC, DNN, SST and SD for the subscriber to configured

Create Subscriber

IMSI* 208930000000001 MSISDN* 9000000001

Key* 465B5CE8B199B49FAA5F0A2EE238A6BC

OPC* E8ED289DEBA952E4283B54E88E6183CA

DNN* internet SST* 1 SD* 000001

Close Submit

Add Subscriber Config

Once the subscriber is configured it will be visible in Testbed UI.

S No	IMSI	Key	OPC	DNN	SST	SD	MSISDN	Actions
1	208930000000002	465B5CE8B199B49FAA5F0A2EE238A6BC	E8ED289DEBA952E4283B54E88E6183CA	internet	1	-	9876543205	  
2	208930000000003	465B5CE8B199B49FAA5F0A2EE238A6BC	E8ED289DEBA952E4283B54E88E6183CA	internet	1	000002		  
3	208930000000004	465B5CE8B199B49FAA5F0A2EE238A6BC	E8ED289DEBA952E4283B54E88E6183CA	internet	1	000001	1234455678	  

Option-2: “Add Bulk Subscribers”

- Useful for adding subscribers in bulk.
- Users need to specify the starting IMSI number and number of subscribers to configured.

Add Bulk Subscriber

IMSI* 208930000032001 No. of UE's* 10

Key* 465B5CE8B199B49FAA5F0A2EE238A6BC

OPC* E8ED289DEBA952E4283B54E88E6183CA

DNN* internet

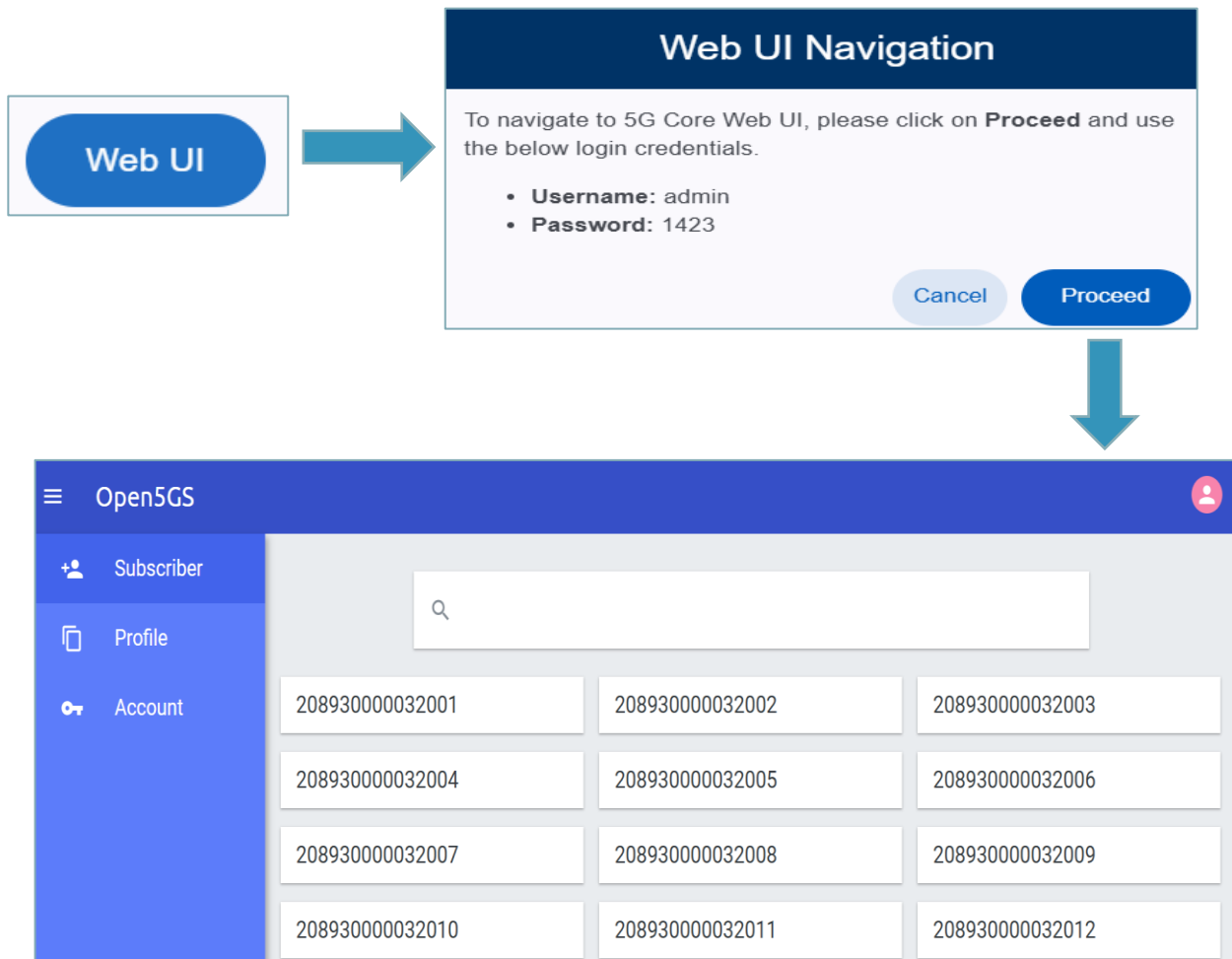
SST* 1 SD* 000001

Close Submit

Add Bulk Subscriber

Option-3: WebUI

- Users can add, delete and modify subscribers one at a time.



4.3.3.3 Logging

This section lets users view or download the logs of a particular 5G core network function.

Configuration
Logging

NF Type*
smf
Since (1 to 1000)*
1
Unit*
hour
View logs
Download Logs

View Logs

To view logs, please select the required **NF Type** and filtering criteria.

4.3.3.4 Call Flows

This section lets users view or download the sample pcap captures for a particular 5G core network function.

Overview

Configuration

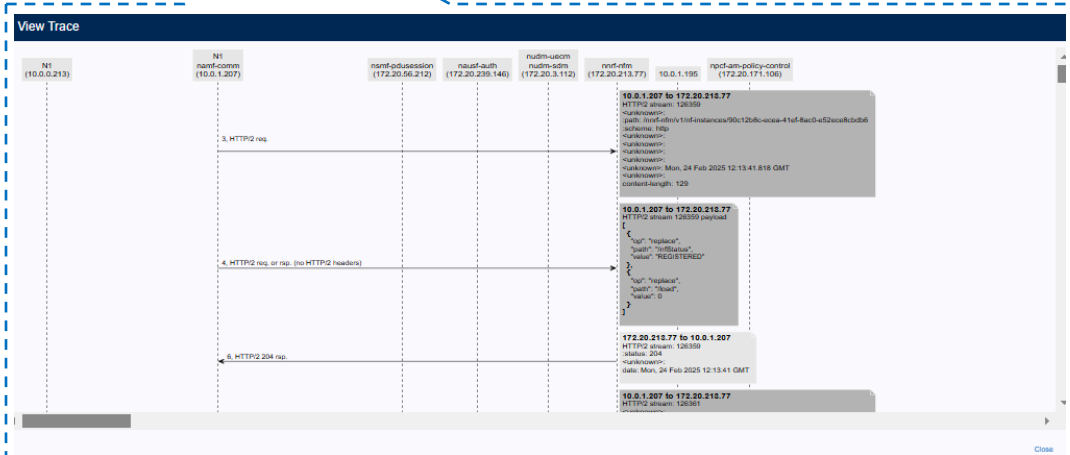
Logging

Call Flows

Sample Call Flows for 5G Core Network Functions

	AMF	SMF	UPF	NRF	NSSF	PCF	AUSF	UDM	UDR
Registration & PDU session Establishment	 	 	 	 	 	 	 	 	 

View Trace



4.3.4 Monitoring

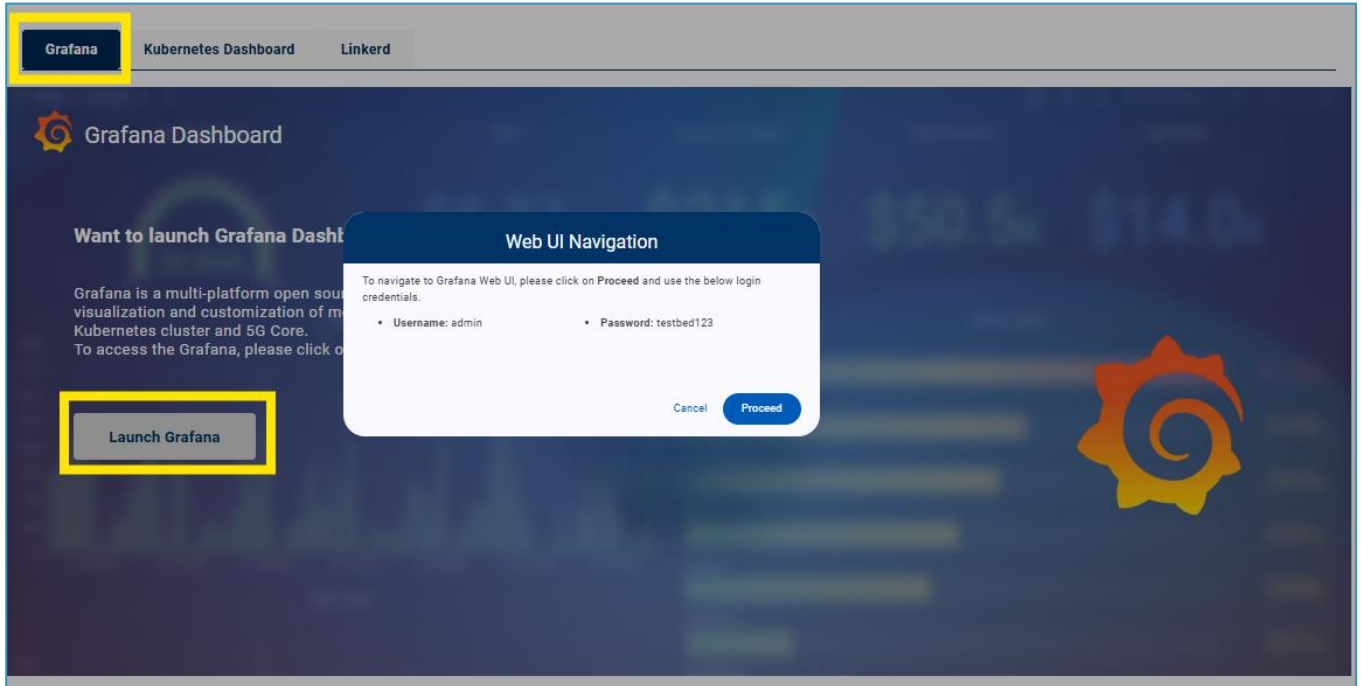
IEEE Testbed provides various tools for monitoring Testbed resources.

- Grafana
- Kubernetes Dashboard
- Linkerd

4.3.4.1 Grafana

Users can view the 5G core metrics by using “Launch Grafana” button. If Grafana is not enabled, users can enable it by using “Enable Grafana” button.

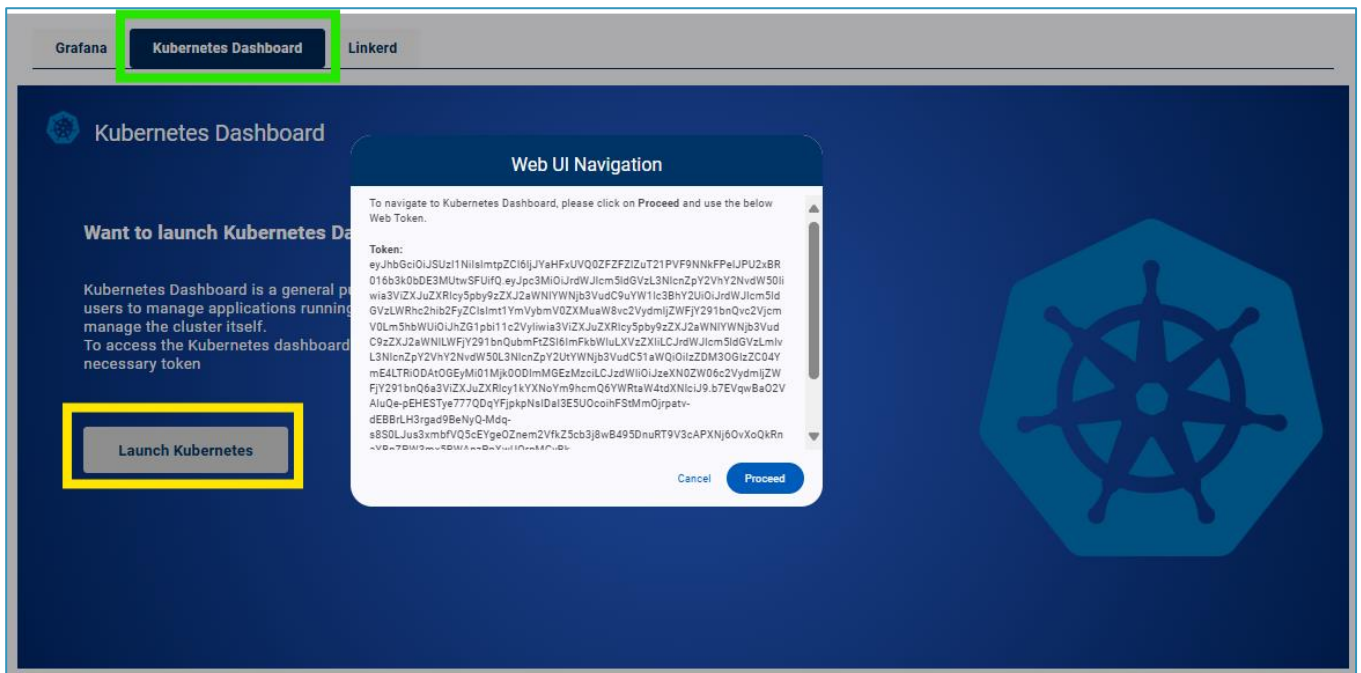
- Username: admin
- Password: testbed123



4.3.4.2 Kubernetes Dashboard

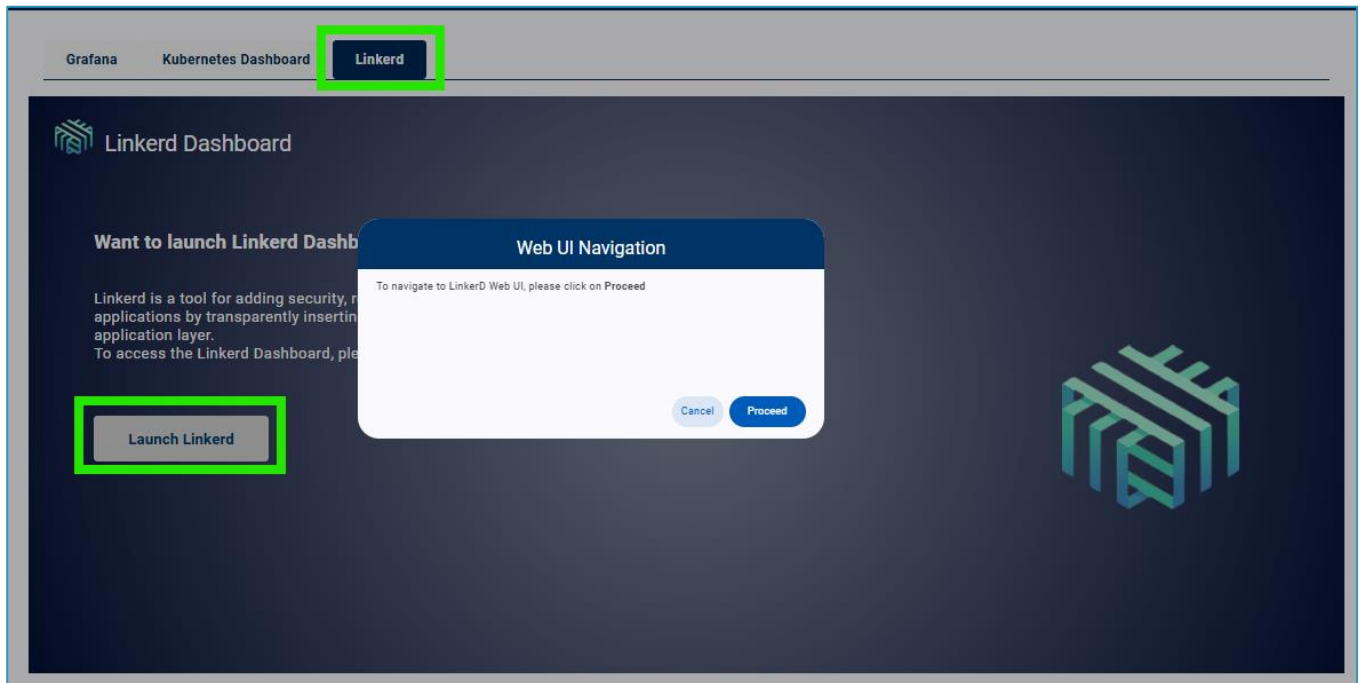
Users can view the detailed Kubernetes dashboard by using “Launch Kubernetes” button. If it's not enabled, user can enable it by using “Enable Kubernetes” button.

Users need to copy the token string and provide it in the Kubernetes window to access the Kubernetes dashboard.



4.3.4.3 Linkerd

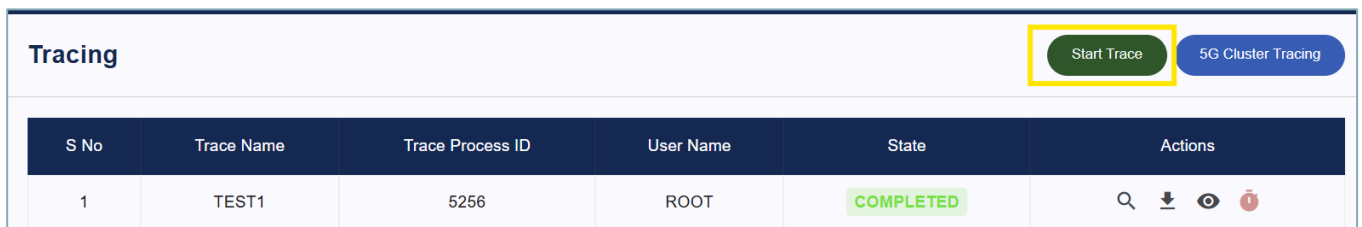
Users can view the detailed Linkerd service mesh by using “Launch Linkerd” button.



4.3.5 Tracing

This section lets the user capture the PCAP traces for their testing.

To start tracing, user need to press “Start trace” button and provide a name.



Start Trace

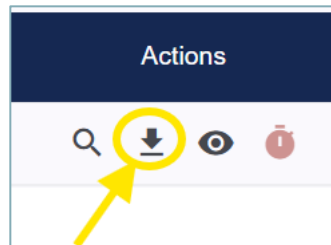
Trace Name*
Test2

Cancel
Start Trace

Once testing is complete, user need to stop the tracing by pressing “timer” button.

Tracing						Note: Trace is running now.Cannot be able to start new trace.	Start Trace	5G Cluster Tracing
S No	Trace Name	Trace Process ID	User Name	State	Actions			
1	TEST2	13036	ROOT	STARTED	Q	↓	👁	⌛
2	TEST1	5256	ROOT	COMPLETED	Q	↓	👁	⌛

Users can download PCAP logs by clicking download button under Actions column.



4.4 Use Cases

4.4.1 CAMARA API

This section lets users view information about the CAMARA API and its use cases and integration with IEEE 5G Testbed.



Please refer to the CAMARA API User Guide for more information

4.4.2 Network Slicing

This section lets users deploy and uninstall the pre-configured slices available in the testbed.

Network Slicing								Create New Slice
S No	Slice ID	Service Slice Type(SST)	Slice Differentiator	Associated Network Functions	Deployment Status	Config	Actions	
1	default_slice	1	1	ALL	true		Deploy Uninstall Modify Delete	
2	slice2	1	2	SMF,UPF	true		Deploy Uninstall Modify Delete	
3	slice3	1	3	SMF,UPF	false		Deploy Uninstall Modify Delete	
4	slice4	1	4	SMF,UPF	false		Deploy Uninstall Modify Delete	

User can also create new slices in the Testbed, provides its slicing information is already known to the 5G core.

Create New Slice

Create Slice

Config:

Slice ID*

SST*

SD*

MCC*

MNC*

TAC*

Default DNN

Name*

IP Address*

Subnet*

IMS DNN

Name*

IP Address*

Subnet*

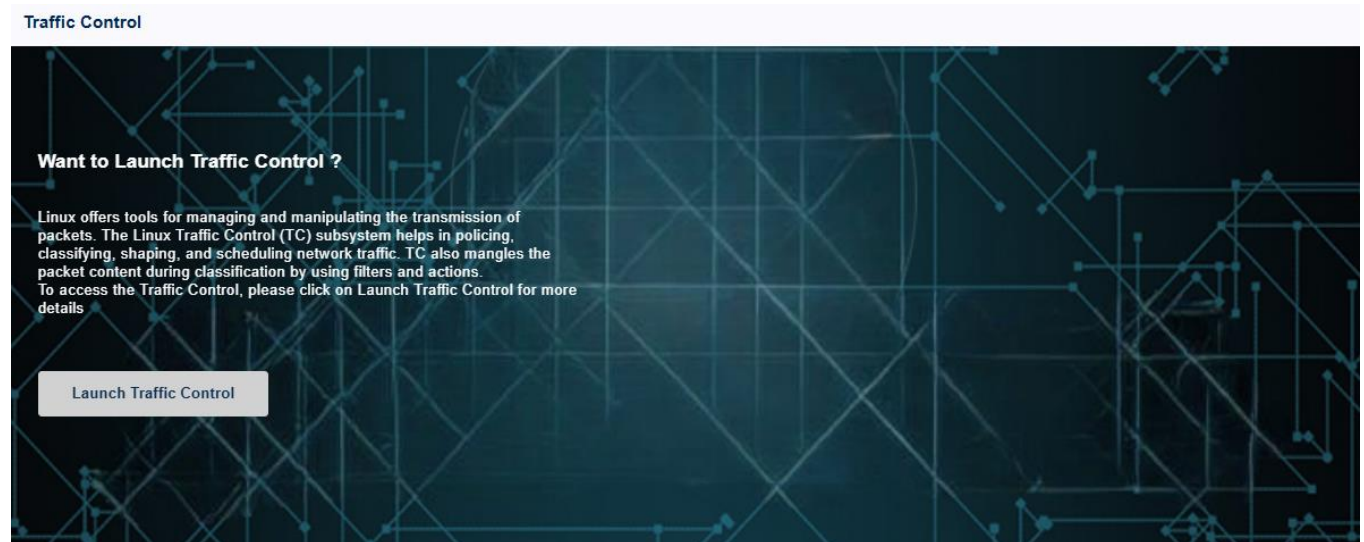
Close

Submit

4.5 Tools

4.5.1 Traffic Control

This section lets users to manage and do manipulation of the transmission of packets using the Linux Traffic Control (TC) subsystem helps in policing, classifying, shaping, and scheduling network traffic.



Once user clicks on “Launch Traffic Control”, user will be redirected to TCGUI for managing and manipulating the transmission of packets.

TCGUI - 3 Available Interfaces

- [docker0](#)
- [ens5](#)
- [lo](#)

docker0

Name	Current Value	New Value	Variance / Correlation	New Value
Rate	None	mbit		
Delay	None	ms	±None	± ms
Loss	None	%	None	%
Duplicate	None	%		
Reorder	None	%	None	%
Corrupt	None	%		
Limit	None			

[Apply docker0 Rules](#)
[Remove docker0 Rules](#)

Launch Traffic Control

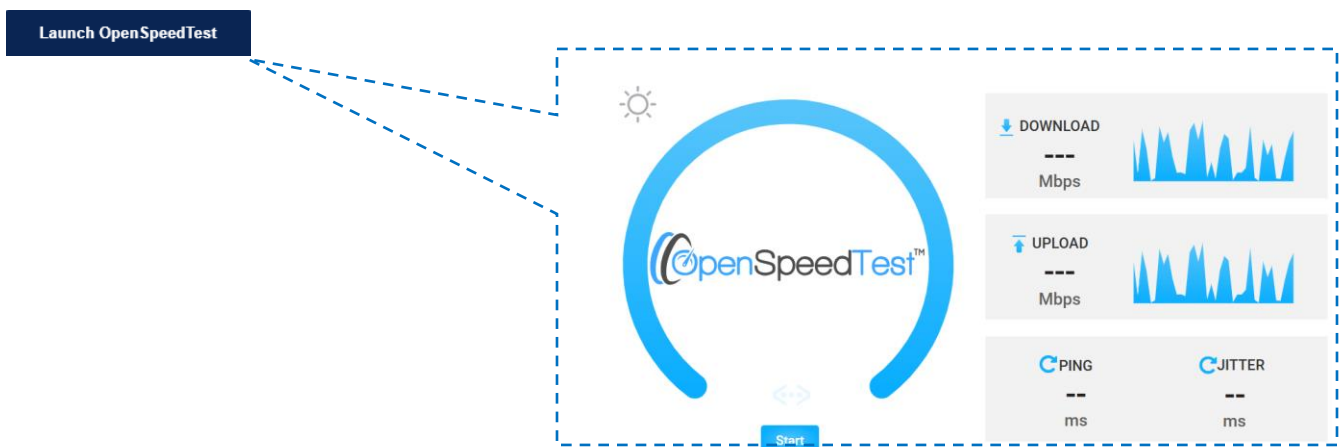
4.5.2 Open Speed Test

This section gives user flexibility to enable, disable and access the Open Speed Test – a Network Performance Estimation and Calculation Tool for data rates and latencies.

Open Speed Test



Once user clicks on “Launch Open Speed Test”, user will be redirected to Open Speed test GUI performing different kind of speed & latency tests



4.5.3 Trace Analyzer

This section gives user flexibility to quickly analyze, investigate 5G SA Network traces and learn 5G with real traces.

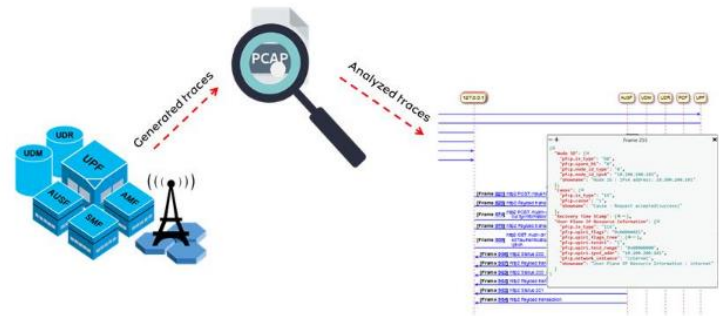
5G Trace Analyzer

Want to Launch Trace Analyzer?

5G Trace & Sequence Analyzer is an open-source project to process and analyze 5G network traces. It lets you Quickly analyze, investigate 5G SA Network traces and learn 5G with real traces.

To access the Trace Analyzer, please click on Launch/Enable Trace Analyzer for more details

[Launch Trace Analyzer](#)



Once user clicks on “Launch Trace Analyzer”, user will be redirected to Trace Analyzer GUI quickly analyze, investigate 5G SA network traces.

[Launch Trace Analyzer](#)



4.5.4 DDoS Attack

This section gives users the ability to perform the NGAP and GTP DDoS (Distributed Denial of Service) attacks on Control and Data planes respectively.

To start a DDoS attack, user need to press “Start DDoS Attack” button and provide the required parameters.

DDoS Attack										Start DDoS Attack
S No	Type of Attack	Source IP Address	Source Port	Dest IP Address	Dest Port	Burst Rate	Duration	Wait Period	State	Actions
1	GTP	10.45.0.19	2152	10.45.0.1	2152	1	1	1	COMPLETED	  

Type of Attack
☒ GTP ☐ NGAP

Burst Rate*
1

Duration (minutes)*
1

Wait Period (seconds)*
1

Source IP Address*
10.45.0.19

Source Port*
2152

Destination IP Address*
10.45.0.1

Destination Port*
2152

Cancel Start Attack

Once the DDoS attack is started, it will be in “RUNNING” state

DDoS Attack										Start DDoS Attack
S No	Type of Attack	Source IP Address	Source Port	Dest IP Address	Dest Port	Burst Rate	Duration	Wait Period	State	Actions
1	GTP	10.45.0.19	2152	10.45.0.1	2152	1	1	1	RUNNING	  
2	NGAP	10.0.1.49	38412	10.0.0.213	30412	3	1	1	COMPLETED	  
3	NGAP	10.0.1.49	38412	10.0.0.213	30412	1	1	1	COMPLETED	  
4	NGAP	10.0.1.49	38412	10.0.0.213	30412	1	1	1	COMPLETED	  
5	NGAP	10.0.1.49	38412	10.0.0.213	30412	1	1	1	COMPLETED	  

Once testing is complete, user need to stop the tracing by pressing “timer” button.

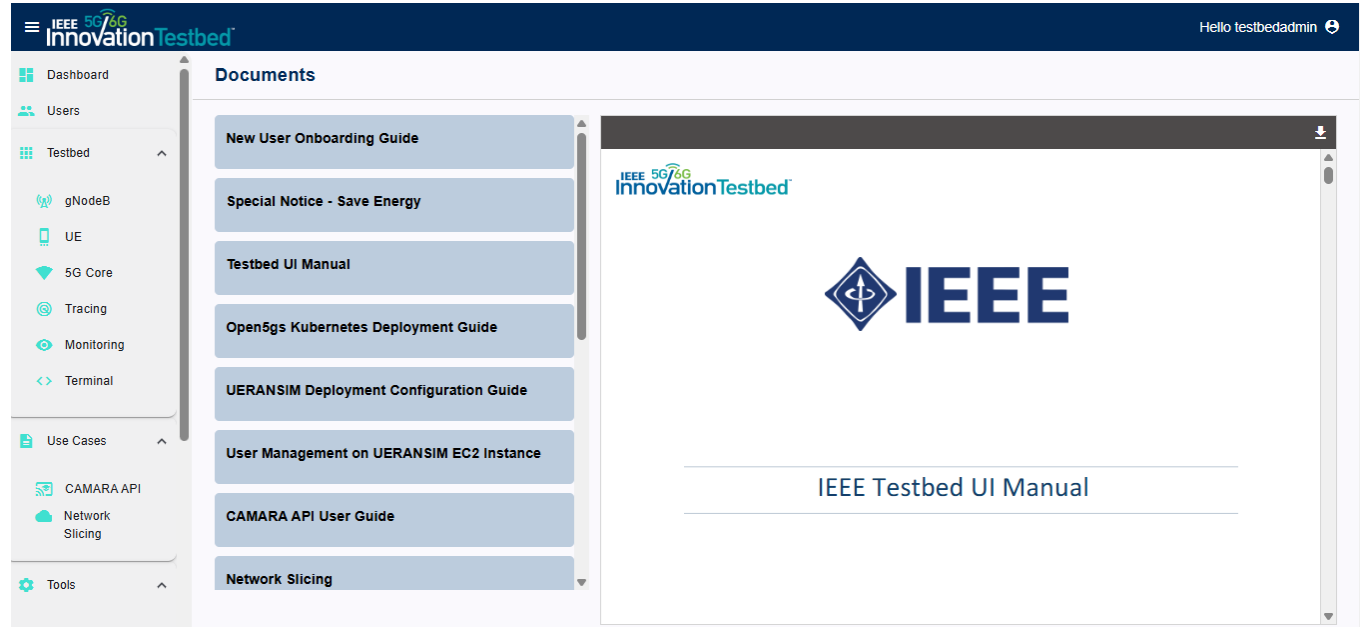
DDoS Attack										Start DDoS Attack
S No	Type of Attack	Source IP Address	Source Port	Dest IP Address	Dest Port	Burst Rate	Duration	Wait Period	State	Actions
1	GTP	10.45.0.19	2152	10.45.0.1	2152	1	1	1	RUNNING	  

4.5 Resources

This section provides resources that include the list of documents and videos for different uses cases and features available in the Testbed.

4.5.5 Documents

This section documents for different uses cases and features available in the Testbed



4.5.6 Videos

This section videos for different uses cases and features available in the Testbed

